Lesson Objective

Interpret signed numbers when used to represent time in situations about speed and direction as well as understand that the product of two negative numbers is positive.

1. Multiply a positive number by a negative number using a number line to determine the sign of the product. Evaluate 4 × (-3) using the number line.  *Answer: -12*	Tell students that 4 × (-3) can be read as 4 groups of -3. Model the first jump from 0 to -3 on the number line, then ask students to complete the remaining three jumps of -3. Ask students what the final position represents and how it relates to the product. Point out that multiplying a positive number by a negative number gives a negative product.
2. Use the distributive property to discover that the product of two negative numbers is positive. Evaluate -4 × (5 + (-2)) two ways. First, evaluate inside the parentheses, then multiply. Then, use the distributive property to rewrite the expression as the sum of two products, and evaluate.	Guide students to first evaluate inside the parentheses: 5 + (-2) = 3. Then multiply -4 × 3 to get -12, using the rule from Problem 1. Next, ask students to use the distributive property to rewrite -4 × (5 + (-2)) as (-4 × 5) + (-4 × (-2)). Have students evaluate -4 × 5 = -20. Ask students what value -4 × (-2) must equal so that -20 + that value equals -12. Guide students to see the product must be positive 8.

Compare the two results to determine the value of -4 × (-2). *Answer: First way: -4 × 3 = -12; Second way: -20 + 8 = -12, so -4 × (-2) = 8*	Tell students that for the distributive property to hold for all rational numbers, a negative times a negative must be positive.
3. Solve multiplication problems with positive and negative rational numbers and summarize the rules for the sign of the product. Evaluate each expression. Then summarize the rules for the sign of the product when multiplying positive and negative numbers. a. 6 × (-7) b. -8 × (-5) c. -20 × (1/10) *Answer: a) -42; b) 40; c) -2; A positive times a positive is positive. A positive times a negative or a negative times a positive is negative. A negative times a negative is positive.*	Students work independently. After students solve, have them state the sign rules in their own words.

Monitor For	**Instructional Moves**
• Do students correctly interpret 4 × (-3) as four jumps of -3 on the number line in Problem 1?	• If a student loses track of jumps on the number line in Problem 1, have them label each jump

• In Problem 2, do students correctly evaluate 5 + (-2) as 3 before multiplying by -4? • In Problem 2, do students reason about the unknown product -4 × (-2) by asking what value, added to -20, gives -12? • In Problem 3c, do students apply the sign rule to -20 × (1/10) and recognize the product is negative, not just compute 20 × (1/10)?	(1st, 2nd, 3rd, 4th) as they go and say "minus 3" with each jump. • In Problem 2, before applying the distributive property, write a(b + c) = ab + ac on the side and ask the student to match each letter to a number in -4 × (5 + (-2)). • If a student says -20 + (-8) = -12 in Problem 2, have them compute -20 + (-8) on a number line to see it gives -28, not -12, and then ask what value they actually need to add to -20. • In Problem 3, ask the student to name the sign of the product before computing the value (e.g., "Will 6 × (-7) be positive or negative? How do you know?").

Reflection Questions

• How did the number line in Problem 1 help you see why 4 × (-3) is negative? *Answer: Each jump of -3 moves left, so four jumps land at -12, which is negative.*

• In Problem 2, how did using the distributive property help you figure out the sign of -4 × (-2)? *Answer: I knew the whole expression equaled -12 and -4 × 5 was -20, so -4 × (-2) had to be positive 8 to make -20 plus that number equal -12.*

• What are the rules for the sign of the product when multiplying positive and negative numbers? *Answer: Positive times positive is positive. Positive times negative or negative times positive is negative. Negative times negative is positive.*

- Why does it make sense that a negative times a negative must be positive? *Answer: If we want the distributive property to keep working with negative numbers, the product of two negatives has to be positive. Otherwise the two ways of evaluating an expression like -4 × (5 + (-2)) wouldn't give the same answer.*